

今更だけどジャンプ加速について

書いてあるものをそのままコピペ

Namespace TASMod(), class TASMod, line 589-592

```
(num10 * num11 / human.mass + vector.y + Physics.gravity.y * Time.fixedDeltaTime)
* (1f - human.rigidbody[0].drag * Time.fixedDeltaTime)
float num9 = Mathf.Pow(Mathf.Clamp(PF.groundManager.Invoke(human)->groudSpeed.y,
0f, 100f), 1.2f);
float num10 = Mathf.Max(0f, (Mathf.Sqrt(1.5f / Physics.gravity.magnitude)
+ num9 / Physics.gravity.magnitude) * human.weight - momentum.y);
float num11 = PF.groundManager.Invoke(human)->groundObjects.Any((GameObject
obj) => PF.grabManager.Invoke(human)->IsGrabbed(obj)) ? 0.75f : 1f;
```

数式で書いてみる

$$v_y = \left(\frac{bc}{105} + v_{y0} + g_y \times \frac{1}{60} \right) \left(1 - 0.05 \times \frac{1}{60} \right)$$
$$a = \begin{cases} 0 & \text{if } v_y \text{ of entire rigidbodies} < 0 \\ (v_y \text{ of entire rigidbodies})^{1.2} & \text{if } 0 \leq v_y \text{ of entire rigidbodies} \leq 100 \\ 100^{1.2} \approx 251.1886431510 & \text{if } 100 < v_y \text{ of entire rigidbodies} \end{cases}$$
$$b = \max \left(0, \left(\sqrt{\frac{1.5}{|\mathbf{g}|}} + \frac{a}{|\mathbf{g}|} \right) \times 105 \times (-9.81) - \sum_{i \in \text{Rigidbody in human}} (v_{yi} \times m_i) \right)$$
$$c = \begin{cases} 0.75 & \text{if human is on grabbed object} \\ 1 & \text{otherwise} \end{cases}$$